1.

Exercise 3

Define a convolutional layer with 3 input channels, 2 output channels, 2×2 kernel. Define a 3×3 input for each input channel, the kernel, and the bias, and compute the output of the layer. Specify the total number of real parameters (weights and biases) for the layer.

Solution:

- we have the following formula:

$$[\mathsf{H}]_{i,j,d} = [\mathbf{u}]_d + \sum_{a=-\Delta}^{\Delta} \sum_{b=-\Delta}^{\Delta} \sum_{c} [\mathsf{V}]_{a,b,c,d} [\mathsf{X}]_{i+a,j+b,c},$$

where $\mathbf{u} \in \mathbb{R}^{c_{out} \times 1}$, $\mathsf{X} \in \mathbb{R}^{c_{in} \times n_h \times n_w}$, $\mathsf{V} \in \mathbb{R}^{c_{out} \times c_{in} \times k_h \times k_w}$, $\mathsf{H} \in \mathbb{R}^{c_{out} \times n'_h \times n'_w}$, and

$$n'_h \times n'_w = (n_h - k_h + 1) \times (n_w - k_w + 1).$$

Input: $n_h = 3$, $n_w = 3$ (3×3 input)

$c_{in} = 3$ input channels

$c_{out} = 2$ output channels

$k_h = 2$, $k_w = 2$ (2×2 kernel)

Ne va da $n'_h \times n'_w = (3 - 2 + 1) \times (3 - 2 + 1) = 2 \times 2$

nr. parametrii $= \underbrace{c_{in} \cdot c_{out} \cdot n'_h \cdot n'_w}_{\text{weights}} + \underbrace{c_{out} \cdot 1}_{\text{bias}} =$

$= 3 \cdot 2 \cdot 2 \cdot 2 + 2 \cdot 1 = 24 + 2 = 26$

Alegem

$$\mathsf{X} = \begin{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \end{bmatrix} \in \mathbb{R}^{3 \times 3 \times 3},$$

$$\mathsf{V} = \underbrace{\begin{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \end{bmatrix}}_{\text{ch 1 out}}, \begin{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}, \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}, \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \end{bmatrix} \in \mathbb{R}^{2 \times 3 \times 2 \times 2}, \rightarrow \text{weights}$$

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^{2 \times 1}. \rightarrow \text{bias}$$

Aplicăm formula pt fiecare ch output.

$$[\mathsf{H}]_{:,:,1} \quad (\text{ch out } 1) = \begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} +$$

$$+ \begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix} \cdot \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} + (1) \leftarrow \text{Bias } u_1 =$$

$$= \begin{pmatrix} 19 & 25 \\ 37 & 43 \end{pmatrix} + \begin{pmatrix} 37 & 47 \\ 67 & 77 \end{pmatrix} + \begin{pmatrix} 63 & 77 \\ 105 & 119 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} =$$

$$= \begin{pmatrix} 120 & 150 \\ 210 & 240 \end{pmatrix} \in \mathbb{R}^{2\times 2}$$

$$[H]_{:,:,2} = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} * \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$
$$+ \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} * \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} + [2] \rightarrow bias_{21}$$
$$= \begin{bmatrix} 43 & 61 \\ 97 & 115 \end{bmatrix} + \begin{bmatrix} 73 & 95 \\ 139 & 161 \end{bmatrix} + \begin{bmatrix} 111 & 137 \\ 189 & 215 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$
$$= \begin{bmatrix} 229 & 295 \\ 427 & 493 \end{bmatrix} \in \mathbb{R}^{2\times 2},$$

$$H = \left(\begin{pmatrix} 120 & 150 \\ 210 & 240 \end{pmatrix}, \begin{pmatrix} 229 & 295 \\ 427 & 493 \end{pmatrix} \right) \in \mathbb{R}^{2\times 2\times 2}$$

Exercițiu curs S11

(2.)

$$X = \left(\begin{pmatrix} 0 & 1 & 0 & -1 & 1 \\ 1 & 0 & 1 & 1 & -1 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & -1 & 0 \end{pmatrix} \right) \in \mathbb{R}^{2\times 5\times 5}$$

padding = 2 , stride = 2 $\Rightarrow p_h = p_w = p = 2$

$\Rightarrow s_h = s_w = s = 2$

$$V = \left(\left(\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \right), \left(\begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix} \right), \left(\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \right) \right)$$

$V \in \mathbb{R}^{3 \times 2 \times 3 \times 3}$

$$U = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^{3 \times 1}$$

$n_h = 5$, $n_w = 5$

$c_{in} = 2$, $c_{out} = 3$

$k_h = 3$, $k_w = 3$ Kernel (se dă)

$$n'_h \times n'_w = \left(\frac{n_h - k_h + p_a + s_a}{s_a} \right) \times \left(\frac{n_w - k_w + p_w + s_w}{s_w} \right) =$$

$$= \left(\frac{5 - 3 + 2 + 2}{2} \right) \times \left(\frac{5 - 3 + 2 + 2}{2} \right) =$$

$$= 3 \times 3$$

$$c_{out} \times n'_h \times n'_w = 3 \times 3 \times 3$$

$\Rightarrow H \in \mathbb{R}$

După padding:

$$X = \left(\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \right)$$

$$H_{\cdot,\cdot,1} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} + \text{element wise}$$

1, 2, 3
4, 5, 6
7, 8, 9

$$= \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} + [1] =$$

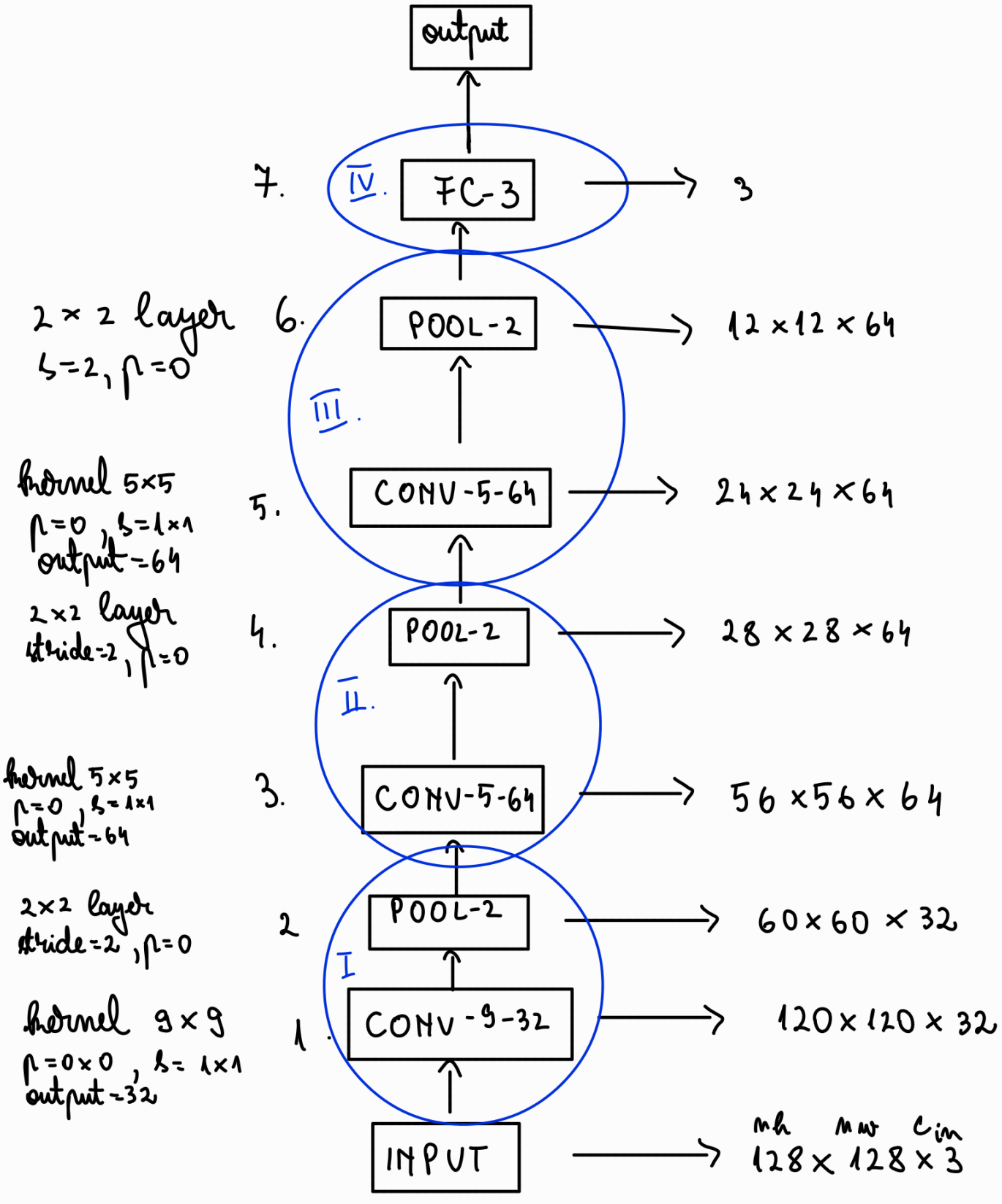
$l=2$
$s=2$

$$= \begin{pmatrix} -1 & 3 & 0 \\ -1 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & -2 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 4 \\ 3 & -1 & 4 \end{pmatrix}$$

la fel si pentru $H_{\cdot,\cdot,2}$, $H_{\cdot,\cdot,3}$

3. Assume we have the following CNN classifier:
INPUT, CONV-9-32, POOL-2, CONV-5-64, POOL-2, CONV-5-64, POOL-2, FC-3.
For each layer, calculate the number of weights, number of biases, and the size of the associated feature maps. The size of INPUT is $128 \times 128 \times 3$. The notation follows the convention:

- CONV-K-N denotes a convolutional layer with N filters, each of them of size K×K, padding and stride parameters are always 0 and 1, respectively.
- POOL-K indicates a K×K pooling layer with stride K and padding 0.
- FC-N stands for a fully-connected layer with N neurons.



$n_h = 128 \qquad c_{in} = 3$

$n_w = 128$

1 $\longrightarrow$ $(128 - 9 + 1) \times (128 - 9 + 1) = 120 \times 120$

parametrii: $\underbrace{32 \cdot 3 \cdot 9 \cdot 9}_{\text{weight}} + \underbrace{32}_{\text{bias}} = 7776 + 32 = 7808$

(c_{out}, c_{in}, kernel, c_{out})

2. $\longrightarrow$ $\left(\frac{120 - 2 + 1}{1}\right) \times \left(\frac{120 - 2 + 1}{1}\right) = 60 \times 60$

nu are parametrii antrenabili

3. $\longrightarrow$ $(60 - 5 + 1) \times (60 - 5 + 1) = 56 \times 56$

(w, b)

$\Omega = 64 \cdot 32 \cdot 5 \cdot 5 + 64 = 51\,200 + 64 = 51\,264$

4 $\rightarrow$ $\left(\frac{56 - 2 + 2}{2}\right) \times \left(\frac{56 - 2 + 2}{2}\right) = 28 \times 28$

5 $\rightarrow$ $(28 - 5 + 1) \times (28 - 5 + 1) = 24 \times 24$

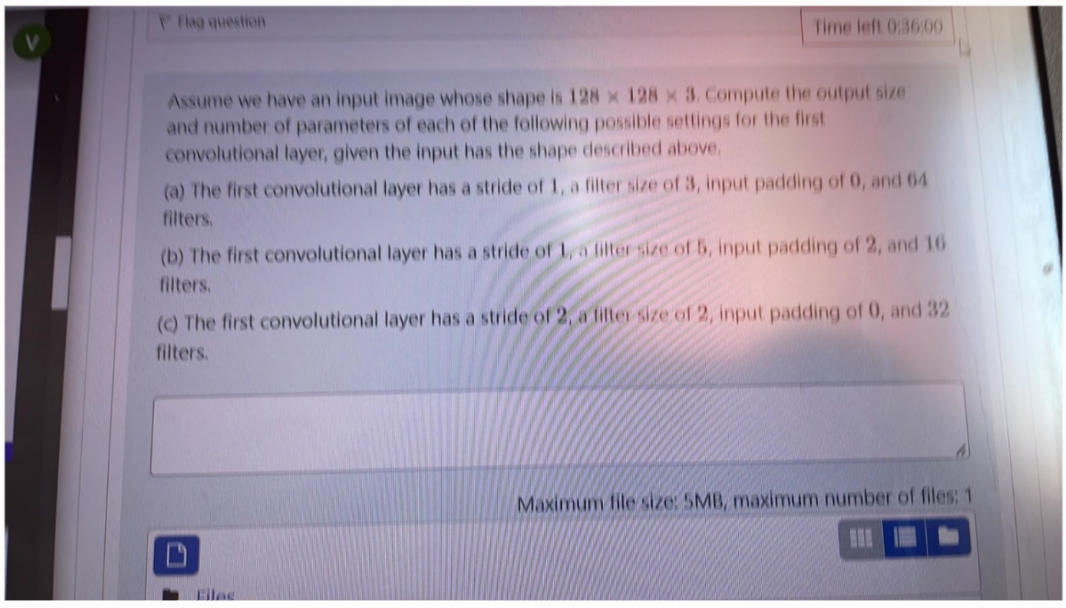
(w, b)

$\Omega = 64 \times 64 \times 5 \times 5 + 64 = 102\,400 + 64 = 102\,464$

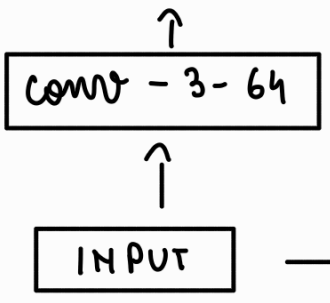
6 $\rightarrow$ $\frac{24 - 2 + 2}{2} \times \frac{24 - 2 + 2}{2} = 12 \times 12$

7 $\rightarrow$ $\Omega = 3 \cdot 64 \cdot 12 \cdot 12 + 3 = 27\,648 + 3 = 27\,651$

4.



a) $n_h = n_w = 128 \quad c_{in} = 3$



$\rightarrow 126 \times 126 \times 64$

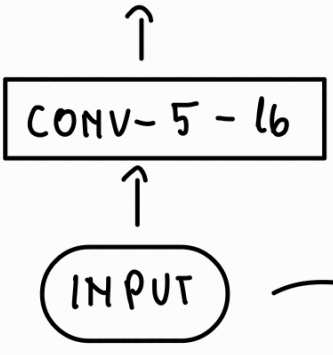
$\rightarrow 128 \times 128 \times 3$

h 3×3
$s = 1 \times 1, p = 0$
$c_{out} = 64$

$$(128 - 3 + 1) \times (128 - 3 + 1) = 126 \times 126$$

$$\underset{w}{} \quad \underset{b}{}$$
$$p = 64 \cdot 3 \cdot 3 \cdot 3 + 64 = 1792$$

b)



$128 \times 128 \times 16$

$128 \times 128 \times 3$

kernel 5×5
$p = 2, s = 1 \times 1$

$$(128 - 5 + 4 + 1) \times (128 - 5 + 4 + 1) = 128 \times 128$$

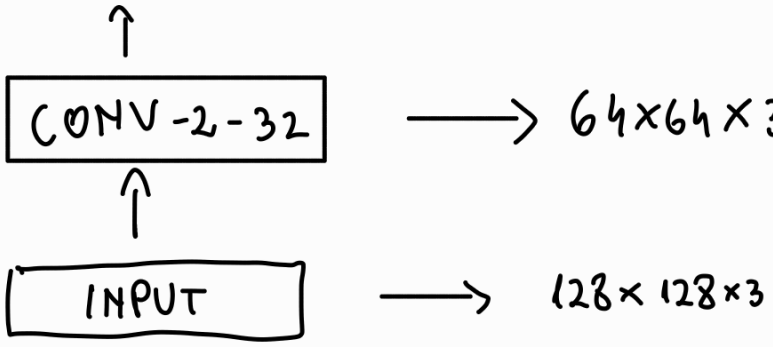
$$\text{param} = 16 \cdot 3 \cdot 5 \cdot 5 + 16 = 1200 + 16 = 1216$$

c)

kernel 2×2

$p = 0$, $s = 2$

$c_{out} = 32$



$\longrightarrow 64 \times 64 \times 32$

$\longrightarrow 128 \times 128 \times 3$

$$\frac{128 - 2 + 0 + 2}{2} \times \frac{128 - 2 + 0 + 2}{2} = 64 \times 64 \quad \frac{n_h - f_h + p + s}{s}$$

$$\text{param} = 32 \cdot 3 \cdot 2 \cdot 2 + 32 = 384 + 32 = 416$$

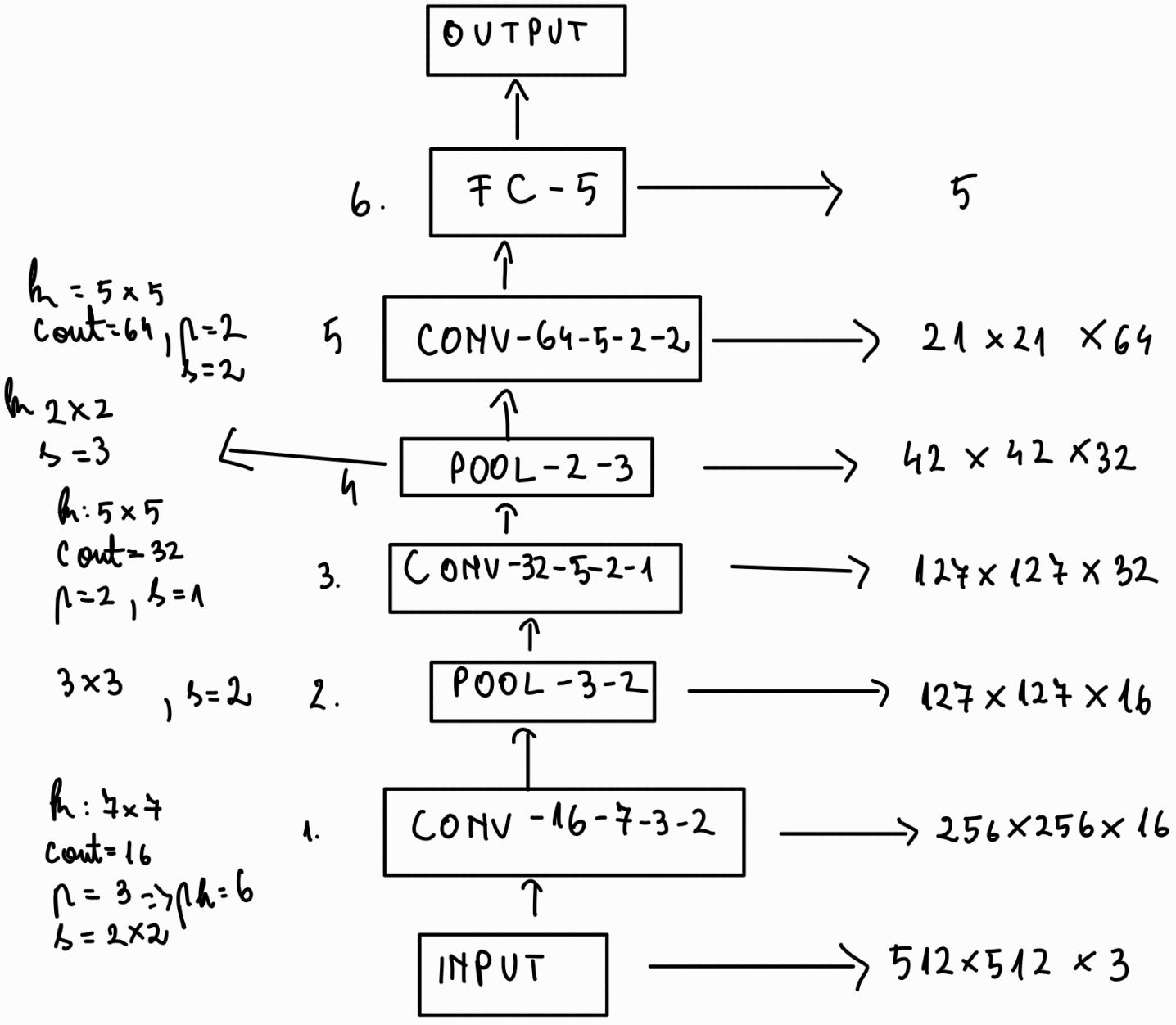
5.

7. Assume we have the following CNN classifier:

INPUT, CONV-16-7-3-2, POOL-3-2, CONV-32-5-2-1, POOL-2-3, CONV-64-5-2-2, FC-5.

For each layer, calculate the number of weights, number of biases, and the shape of the associated feature maps. The shape of INPUT is $512 \times 512 \times 3$. The notation follows the convention:

- CONV-N-K-P-S denotes a K×K convolutional layer with N channels, padding P, and stride S.
- POOL-K-S indicates a K×K pooling layer with stride S and padding 0.
- FC-N stands for a fully-connected layer with N units.



1. $\left\lfloor \frac{512 - 7 + 3 \cdot 2 + 2}{2} \right\rfloor \times \left\lfloor \frac{512 - 7 + 3 \cdot 2 + 2}{2} \right\rfloor = 256 \times 256$

$\text{nparam} = 16 \cdot 3 \cdot 7 \cdot 7 + 16$

2. $\left\lfloor \frac{256 - 3 + 2}{2} \right\rfloor \times \text{-11-} = 127 \times 127$

3. $\left(127 - 5 + 2\cdot 2 + 1\right) \times \quad 127 \quad = 127 \times 127$

$\rho = 32 \cdot 16 \cdot 5 \cdot 5 + 32$

4. $\left\lfloor \frac{127 - 2 + 3}{3} \right\rfloor \times \text{-11-} \quad = 42 \times 42$

5. $\left\lfloor \frac{42 - 5 + 2\cdot 2 + 2}{2} \right\rfloor \times \text{-11-} \quad = 21 \times 21$

$\rho = 64 \cdot 32 \cdot 5 \cdot 5 + 64$

6. $\rho = 5 \cdot 64 \cdot 21 \cdot 21 + 5$

(6.)

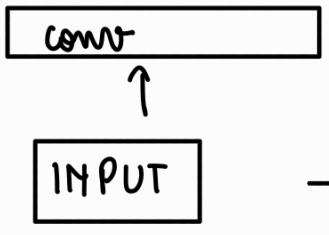
3. Assume we have an input image whose shape is $3 \times 224 \times 224$. Compute the output shape and number of parameters (weights and biases) of each of the following possible settings for the first convolutional layer, given the input has the shape described above.

(a) The first convolutional layer has a stride of 2, a kernel size of 3, input padding of 0, and 64 output channels.

(b) The first convolutional layer has a stride of 4, a kernel size of 5, input padding of 2, and 32 output channels.

a)

$k: 3\times 3$
$p=0,\ s=2$
$c_{out}=64$



$\rightarrow 64 \times 111 \times 111$

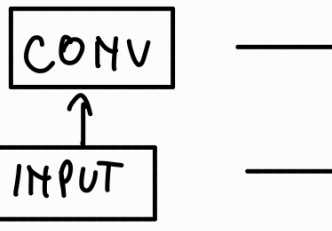
$\rightarrow 3 \times 224 \times 224$

$\left\lfloor \frac{224 - 3 + 0 + 2}{2} \right\rfloor \times \left\lfloor \frac{224 - 3 + 0 + 2}{2} \right\rfloor = 111 \times 111$

$\rho = 64 \cdot 3 \cdot 3 \cdot 3 + 64$

b)

5×5
$p=2,\ s=4$
$c_{out}=32$



$\rightarrow 32 \times 57 \times 57$

$\rightarrow 3 \times 224 \times 224$

$$\left\lfloor \frac{224 - 5 + 2 \cdot 2 + 4}{4} \right\rfloor \times -11- = 57 \times 57$$

$$\Omega = 32 \cdot 3 \cdot 5 \cdot 5 + 32$$